

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-afd90832ef61e41f5.jsonl`  
- SHA-256: `0c500ca420a6a78108a87ea90fe102de7c21aceaf63296b7fd27f24e35548d08`  
- Source modified: `2026-05-31T04:22:04+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.041Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Pixtral-12B-2409 model weights are released under the Apache-2.0 license, making the weights commercially safe (not a restrictive custom/community/non-commercial license)."

****Source:**** <https://huggingface.co/mistralai/Pixtral-12B-2409> (primary)

****Supporting quote:**** "License: apache-2.0"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.493Z)

I'll verify this claim about Pixtral-12B-2409's license. Let me start by loading the search tools and checking the primary source.

3. assistant (2026-05-31T04:21:13.213Z)

I have the search tools. Let me verify the claim by checking the primary HuggingFace source and searching for any contradicting evidence.

4. user (2026-05-31T04:21:17.370Z)

Pixtral-12B-2409 License

Based on the web page content, the Pixtral-12B-2409 model is released under the **Apache 2.0 license**.

Evidence from the page:

- 1. **License metadata tag**: The page displays "License: apache-2.0" in the model information section at the top.
- 2. **Key features section**: States "License: Apache 2.0"

This is an open-source license that allows for broad usage, modification, and distribution of the model with appropriate attribution.

5. user (2026-05-31T04:21:20.146Z)

Web search results for query: "Pixtral-12B-2409 license Apache 2.0 Mistral"

Links: [{"title": "mistralai/Pixtral-12B-2409 · Hugging Face", "url": "https://huggingface.co/mistralai/Pixtral-12B-2409"}, {"title": "README.md · mistralai/Pixtral-12B-2409 at main", "url": "https://huggingface.co/mistralai/Pixtral-12B-2409/blob/main/README.md"}, {"title": "mistralai/Pixtral-12B-Base-2409 · Hugging Face", "url": "https://huggingface.co/mistralai/Pixtral-12B-Base-2409"}, {"title": "Pixtral-12B-2409 is now available on Amazon Bedrock Marketplace | Artificial Intelligence", "url": "https://aws.amazon.com/blogs/machine-learning/pixtral-12b-2409-is-now-available-on-amazon-bedrock-marketplace/"}, {"title": "Pixtral 12B 2409 by Mistral AI on Vercel AI Gateway, Specs, Pricing & API", "url": "https://vercel.com/ai-gateway/models/pixtral-12b"}, {"title": "Pixtral 12B: A Guide With Practical Examples | DataCamp", "url": "https://www.datacamp.com/tutorial/pixtral-12b"}, {"title": "unsloth/Pixtral-12B-2409 · Hugging Face", "url": "https://huggingface.co/unsloth/Pixtral-12B-2409"}, {"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral-12B-Base-2409 huggingface.co api & mistralai Pixtral-12B-Base-2409 github AI Model - Toolify", "url": "https://www.toolify.ai/ai-model/mistralai-pixtral-12b-base-2409"}, {"title": "unsloth/Pixtral-12B-2409-bnb-4bit · Hugging Face", "url": "https://huggingface.co/unsloth/Pixtral-12B-2409-bnb-4bit"}]

Based on the search results, I found comprehensive information about the Pixtral-12B-2409 model's license:

Pixtral-12B-2409 is released under the Apache 2.0 license. Similar to other Mistral models, such as Mistral 7B, Mixtral 8x7B, Mixtral 8x22B, and Mistral Nemo 12B, Pixtral 12B is released under the commercially permissive Apache 2.0, providing enterprise and startup customers with a high-performing VLM option to build complex multimodal applications.

The Pixtral-12B-2409 is a Multimodal Model of 12B parameters plus a 400M parameter vision encoder. Released in September 2024 under Apache 2.0 license, it handles image analysis, OCR, chart interpretation, and visual question answering with a 128,000-token context window.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:23.254Z)

The primary source confirms Apache-2.0. Let me cross-check with independent third-party sources to ensure there's no qualification, hidden restriction, or contradiction ? and verify the raw README metadata directly.

7. user (2026-05-31T04:21:27.684Z)

License Information for Pixtral-12B-2409

YAML Frontmatter License Field

```
```\nlicense: apache-2.0\n```
```

## License Terms and Usage Restrictions

The model is released under the **Apache 2.0 license**.

### Notable Limitation Mentioned in Body:

The documentation includes an important limitation statement:

> "The Pixtral model does not have any moderation mechanisms. We're looking forward to engaging with the community on ways to make the model finely respect guardrails, allowing for deployment in environments requiring moderated outputs."

This indicates that while the model is Apache 2.0 licensed (permissive), users should be aware that there are no built-in content moderation safeguards, which may be relevant for deployment in production environments that require moderated outputs.

**8. user (2026-05-31T04:21:28.364Z)**

## Pixtral 12B License

According to Mistral's official announcement, Pixtral 12B is released under the **Apache 2.0 license**.

The exact wording states: "License: Apache 2.0"

This is listed in the "Use" section of the model's key characteristics on the announcement page.

**9. assistant (2026-05-31T04:21:32.270Z)**

I now have convergent confirmation from multiple sources. Let me do one final check for any contradicting evidence or controversy about Mistral relicensing or restrictions on this specific model.

**10. user (2026-05-31T04:21:38.174Z)**

Web search results for query: "Pixtral 12B license change restriction non-commercial MRL Mistral Research License"

Links: [{"title": "[Deprecated] Pixtral Large | Mistral AI", "url": "https://mistral.ai/news/pixtral-large/"}, {"title": "GitHub - mistralai/mistral-inference: Official inference library for Mistral models", "url": "https://github.com/mistralai/mistral-inference"}, {"title": "AI in abundance | Mistral AI", "url": "https://mistral.ai/news/september-24-release/"}, {"title": "mistral-community/pixtral-12b-240910 - License?", "url": "https://huggingface.co/mistral-community/pixtral-12b-240910/discussions/7"}, {"title": "New Mistral Small 3 does more with less under Apache license", "url": "https://the-decoder.com/new-mistral-small-3-does-more-with-less-under-apache-license/"}, {"title": "Changelog | Mistral Docs", "url": "https://docs.mistral.ai/resources/changelogs"}, {"title": "Add MRL license - zachlandes/Mistral-Large-Instruct-2411-MLX at 4dd8a8e", "url": "https://huggingface.co/zachlandes/Mistral-Large-Instruct-2411-MLX/commit/4dd8a8e698cae04965fad9e24347d162bbdbc6bf"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}, {"title": "Mistral", "url": "https://mistral.ai/static/licenses/MRL-0.1.md"}]

Based on the search results, here's what I found regarding Pixtral 12B's license:

Pixtral 12B is available under the Apache 2.0 license, which is a permissive open-source license. This is actually notable because it represents a departure from the Mistral Research

License (MRL) approach used by some other Mistral models.

The MRL itself is a proprietary license with specific restrictions. The MRL restricts use to Research Purposes, and for any purpose not expressly authorized under the MRL agreement, users must request a license from Mistral AI.

However, Pixtral 12B specifically does **not** use this restrictive license. Pixtral 12B is released under the Apache 2.0 license, which is much more permissive. Mistral moved away from its proprietary MRL to the more permissive Apache 2.0 license, which allows users to freely use, modify, and redistribute the models, even for commercial purposes.

So to clarify: Pixtral 12B is **not** restricted by the MRL's non-commercial limitation? it uses the fully permissive Apache 2.0 license instead.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 11. assistant (2026-05-31T04:21:51.460Z)

The evidence is convergent and the claim is well-supported. Let me summarize my findings.

**Verification of the claim:** "Pixtral-12B-2409 model weights are released under the Apache-2.0 license, making the weights commercially safe."

**Checklist results:**

- Supported by quote?** Yes. The YAML frontmatter metadata reads verbatim `license: apache-2.0`, confirming the supporting quote is the actual model-card metadata, not an inference.
- Contradicting evidence?** None found. Multiple independent sources confirm Apache-2.0. Critically, the research question warns that "several VLM weights ship custom restrictive licenses" ? so I specifically searched whether Pixtral falls under Mistral's restrictive MRL (Mistral Research License, non-commercial/research-only). The result is the opposite: Pixtral 12B is explicitly contrasted *against* MRL. Sources note it is "released under the commercially permissive Apache 2.0," consistent with Mistral's other Apache-2.0 models (Mistral 7B, Mixtral, Nemo). This is the correct skeptical angle to check and it confirms rather than refutes.
- Source quality sufficient?** Yes. Primary source is Mistral's own HuggingFace repo (both rendered card and raw README YAML), plus Mistral's official announcement page. These are authoritative first-party sources for a license claim.
- Outdated?** The model dates to September 2024. The claim is about the original release license, which Apache-2.0 grants irrevocably ? once weights are released under Apache-2.0, that grant cannot be retroactively revoked for those weights. No evidence of relicensing. The claim is current (confirmed via 2026-dated third-party sources).
- Marketing/cherry-picked?** No. License metadata is a factual repo field, not a benchmark or marketing claim.

One nuance worth noting for the broader research (does not affect this claim): the Apache-2.0 grant applies to **Pixtral-12B-2409** specifically. Mistral's *other* vision model, **Pixtral Large**, uses the restrictive MRL ? so the per-model license discipline the research question demands is real and Pixtral-12B-2409 is on the safe side of it. The 12B weights themselves carry no acceptable-use restrictions (only a noted absence of content moderation, which is not a license restriction).

The claim is well-supported, current, and backed by primary first-party sources. I could not refute it.

### 12. user (2026-05-31T04:21:58.288Z)

Structured output provided successfully

### 13. assistant (2026-05-31T04:22:04.211Z)

Verification complete. The claim that Pixtral-12B-2409 weights are Apache-2.0 licensed and commercially safe is **\*\*well-supported and could not be refuted\*\*** (refuted=false, high confidence).

Key findings:

- The raw HuggingFace README YAML frontmatter literally reads `license: apache-2.0` ? the supporting quote is the actual metadata, not an overreach.
- Corroborated by Mistral's official announcement and multiple independent third-party sources.
- The critical skeptical test (the research question warns many VLM weights carry restrictive custom/non-commercial licenses) actually strengthens the claim: Pixtral-12B-2409 is explicitly contrasted *against* Mistral's restrictive MRL and is part of Mistral's Apache-2.0 line.
- Important per-model caveat for the broader research: Mistral's *Pixtral Large* uses the restrictive MRL, so license must be verified per-model ? but Pixtral-12B-**\*\*2409\*\*** specifically is on the safe side.